

UPPER EXTREMITY PATTERN 1



Test Objective for the Upper Extremity Pattern 1:

Upper Extremity Pattern 1 assesses shoulder internal rotation, shoulder extension, elbow flexion, and thorax extension/rotation.

How to Perform the Upper Extremity Pattern 1:

The patient assumes the starting position by standing erect with feet together and toes pointing forward. The person then reaches back with the right arm, trying to touch the inferior angle of the left scapula. Repeat the motion with the left arm.

Functional Criteria:

- Touches inferior angle of opposite scapula
- Without excessive effort and/or loss of motor control

Common Compensations:

- Radial Deviation
- Scapular winging
- Two movements rather than one fluid movement

Tips for Testing:

- Observe from the back and side views.
- Head position should remain unchanged throughout the movement.
- Do not coach the movement; simply repeat the instructions if needed.
- Was there pain?



UPPER EXTREMITY PATTERN 2



Test Objective for the Upper Extremity Pattern 2:

Upper Extremity Pattern 2 assesses shoulder external rotation, shoulder flexion and abduction, elbow flexion, and thorax extension/rotation.

How to Perform the Upper Extremity Pattern 2:

The patient assumes the starting position by standing erect with feet together and toes pointing forward. The person then reaches overhead with the right arm in a “combing hair” motion, trying to touch the spine of the left scapula. Repeat the motion with the left arm.

Functional Criteria:

- Touches spine of the opposite scapula
- Without excessive effort and/or loss of motor control

Common Compensations:

- Torso rotation
- Cervical flexion and/or rotation
- Opposite shoulder elevation

Tips for Testing:

- Observe from the back and side views.
- Head position should remain unchanged throughout the movement.
- Do not coach the movement; simply repeat the instructions if needed.
- Was there pain?



UPPER EXTREMITY BREAKOUTS



Upper Extremity Pattern Breakouts Rationale

Thorax Mobility Dysfunction vs. Thorax Stability and/or Motor Control Dysfunction

A clinical pearl within the SFMA is a Thorax Mobility Dysfunction finding is a priority over Shoulder Dysfunction findings. Therefore, the Upper Extremity Pattern Breakouts begin by assessing bilateral Thorax extension/rotation mobility and/or stability/motor control using the Lumbar Locked Extension/Rotation Test, actively and then followed by passively if needed. No matter your finding with Lumbar Locked Extension/Rotation Test, you will continue to look at the shoulder girdle for more information.

Shoulder Girdle Pattern Dysfunction vs Pattern Parts Dysfunction

Our first step in assessing the shoulder girdle will be to reduce the postural stability requirements by having the patient lie prone with the head supported on a treatment table. In this position, the thoracic spine, shoulder girdle and cervical spine have minimal stability requirements. Next, instruct the patient to try to repeat the top tier shoulder patterns as previously described.

If the Active Prone Upper Extremity Pattern Test is now functional and non-painful, and the thorax was cleared for mobility, the patient has a postural and/or a shoulder girdle stability/motor control dysfunction that is affecting the upper extremity pattern. If the Active Prone Upper Extremity Pattern Test is functional but the thorax has a mobility dysfunction, the shoulder itself has sufficient mobility and the thorax should be treated. If the Active Prone Upper Extremity Pattern Test is dysfunctional and/or painful, check the same motion passively.

If the Passive Prone Upper Extremity Pattern Test is now functional and non-painful, the patient has a postural and/or shoulder girdle stability/motor control dysfunction. Remember, if there is a thorax mobility finding from the Lumbar Locked Extension/Rotation Test, that is the priority finding. If passive pattern motion is still dysfunctional and/or painful, it is appropriate to check the parts of each pattern separately.

For Upper Extremity Pattern 1, the parts to assess are shoulder internal rotation, shoulder extension, and elbow flexion with the arm extended. Check active and passive for each movement to differentiate between mobility and stability/motor control dysfunctions if needed.

For Upper Extremity Pattern 2, the parts to assess are shoulder external rotation, shoulder flexion/abduction, and elbow flexion with the arm flexed. Check active and passive for each movement to differentiate between mobility and stability/motor control dysfunctions if needed.

If there is pain with any passive testing of the parts, note where the pain is and come back and examine the region in more detail after completing the entire SFMA.

UPPER EXTREMITY PATTERN 1 BREAKOUT



Selective Breakout Tests for Upper Extremity Pattern 1 Breakout:



Active Lumbar Locked
Extension/Rotation Test



Passive Lumbar Locked
Extension/Rotation Test



Active Prone Upper
Extremity Pattern 1



Passive Prone Upper
Extremity Pattern 1



Active Prone Shoulder
90/90 IR Test



Passive Prone Shoulder
90/90 IR Test



Active Prone Shoulder
Extension Test



Passive Prone Shoulder
Extension Test



Active Prone Elbow
Flexion Test



Passive Prone Elbow
Flexion Test

UPPER EXTREMITY PATTERN 2 BREAKOUT



Selective Breakout Tests for Upper Extremity Pattern 2 Breakout:



Active Lumbar Locked
Extension/Rotation Test



Passive Lumbar Locked
Extension/Rotation Test



Active Prone Upper
Extremity Pattern 2



Passive Prone Upper
Extremity Pattern 2



Active Prone Shoulder
90/90 ER Test



Passive Prone Shoulder
90/90 ER Test



Active Prone Shoulder
Abduction Test



Passive Prone Shoulder
Abduction Test

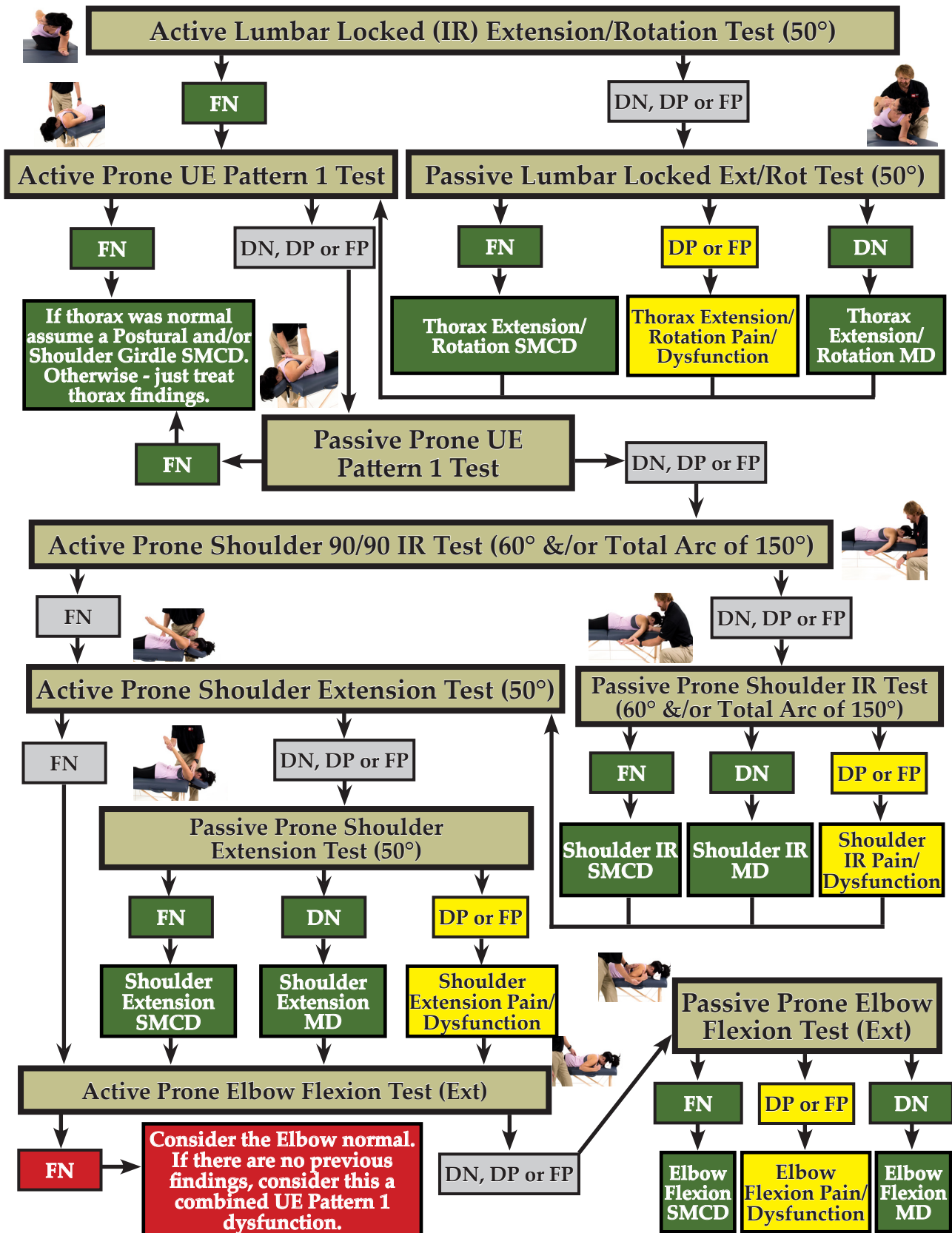


Active Prone Elbow
Flexion Test

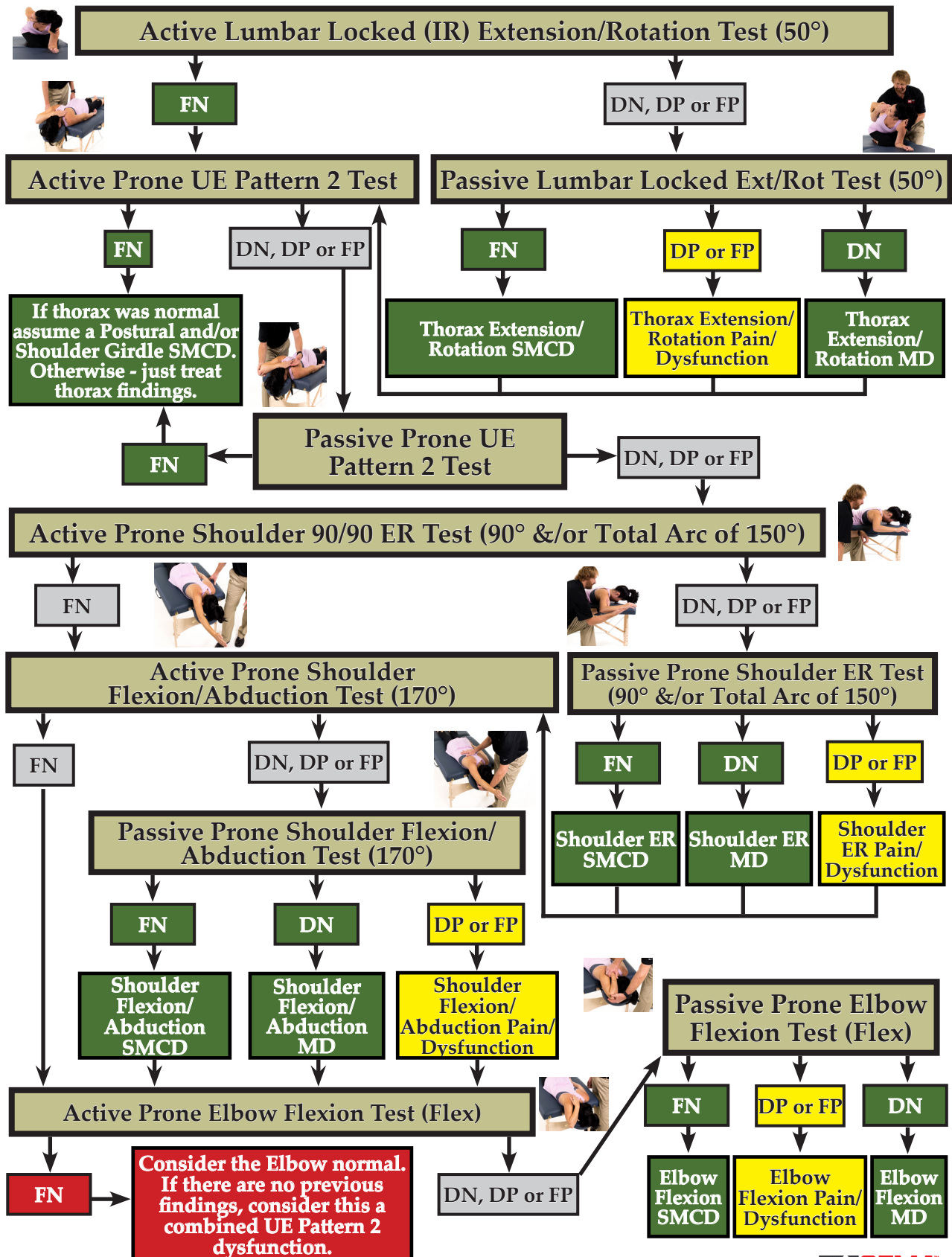


Passive Prone Elbow
Flexion Test

UPPER EXTREMITY PATTERN 1 BREAKOUT



UPPER EXTREMITY PATTERN 2 BREAKOUT



UPPER EXTREMITY BREAKOUTS



Test Objective for the Active Lumbar Locked (IR) Extension/Rotation Test:

To test the combined active extension and rotation in the thorax in an unloaded posture.

How to Perform the Active Lumbar Locked (IR) Extension/Rotation Test:

Instruct the patient to get into a full prone rocking position (butt to heels) with their forearms together in front of knees. Their knees do not need to be together. Place their right hand and forearm behind on the small of their back with their palm facing upwards. To set their scapula in proper position, the shoulder should be in a mid-axial position. Their left forearm and hand remain on the testing surface in the start position. Have the person rotate their head and thorax to the right as far as possible while maintaining the set-up position. Compare both sides.

Functional Criteria:

When using a line through the AC Joints as the reference line, normal thorax extension/rotation is 50 degrees relative to the horizontal bilaterally, without lateral flexion/side bending.

Additional Information:

The prone rocking position helps eliminate the lumbar spine from contributing to the extension/rotation. This can also differentiate between right and left thorax restrictions.

If the patient has pain with the shoulder IR test position, it is appropriate to use a chest position for the arm - their right arm across their chest, resting right hand on their left shoulder.

Possible Findings:

- FN: 50 degrees or greater of thorax extension/rotation, no pain
 - » Consider Thorax Extension/Rotation normal.
 - Proceed to the Active Prone UE Pattern 1 Test.
- FP: 50 degrees or greater of thorax extension/rotation, pain
 - » Proceed to Passive Lumbar Locked (IR) Extension/Rotation test.
- DN: Less than 50 degrees of thorax extension/rotation either on one side (unilateral DN), or on both sides (bilateral DN), no pain
 - » Proceed to Passive Lumbar Locked (IR) Extension/Rotation test.
- DP: Less than 50 degrees of thorax extension/rotation either on one side (unilateral DP), or on both sides (bilateral DP), pain
 - » Proceed to Passive Lumbar Locked (IR) Extension/Rotation test.



UPPER EXTREMITY BREAKOUTS



Test Objective for the Passive Lumbar Locked (IR) Extension/Rotation Test:

To test the combined passive extension and rotation of the thorax in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Lumbar Locked (IR) Extension/Rotation Test:

Instruct the patient to get into a full prone rocking position (butt to heels) with their forearms together in front of knees. Place their right hand and forearm behind on the small of their back with their palm facing upwards. To set their scapula in proper position, the shoulder should be in a mid-axial position. Their left forearm and hand remain on the testing surface in the start position. Stabilize on the outside of their left shoulder with your left hand and at their hip, by placing your right hip to their left hip. This allows you to create a tunnel for rotation. Thread your right arm through their elbow, placing your right forearm tightly along their trunk with ulnar border of your hand over their coracoid process. Passively rotate the thorax to the right as far as possible while maintaining the set-up position. Compare both sides.

Functional Criteria:

When using a line through the AC Joints as the reference line, normal thorax extension/rotation is 50 degrees relative to the horizontal bilaterally, without lateral flexion/side bending.

Additional Information:

The prone rocking position helps eliminate the lumbar spine from contributing to the extension/rotation. This can also differentiate between right and left thoracic restrictions.

If the patient has pain with the shoulder IR test position, it is appropriate to use a chest position for the arm - their right arm across their chest, resting right hand on their left shoulder.

Possible Findings:

- FN: 50 degrees or greater of thorax extension/rotation, no pain
 - » Indicates a thorax extension/rotation stability/motor control dysfunction. Proceed to the Active Prone UE Pattern 1 Test.
- FP: 50 degrees or greater of thorax extension/rotation, pain
 - » Indicates pain in the thorax with extension/rotation - complete local biomechanical exam. Proceed to the Active Prone UE Pattern 1 Test.
- DN: Less than 50 degrees of thorax extension/rotation either on one side (unilateral DN), or on both sides without pain (bilateral DN)
 - » Indicates a thorax extension/rotation mobility dysfunction - complete local biomechanical exam Proceed to the Active Prone UE Pattern 1 Test.
- DP: Less than 50 degrees of thorax extension/rotation either on one side (unilateral DP), or on both sides with pain (bilateral DP)
 - » Indicates a painful thorax extension/rotation mobility dysfunction - complete local biomechanical exam. Proceed to the Active Prone UE Pattern 1 Test.



UPPER EXTREMITY BREAKOUTS



Test Objective for the Active Prone Upper Extremity Pattern 1 Test:

To test active shoulder mobility and stability/motor control in an unloaded posture.

How to Perform the Active Prone Upper Extremity Pattern 1 Test:

Instruct the patient to assume the prone position with the arms and hands by the thighs. Instruct the patient to try to touch their opposite inferior angle of their scapula by reaching up and behind their back, repeating the Upper Extremity Pattern 1 test.

Functional Criteria:

Normal motion is fingertips reaching the opposite inferior angle of the scapula, without excessive scapular winging, without radial deviation, and/or without anterior dumping of the shoulder.

Additional Information:

In this position the thoracic spine, shoulder girdle and cervical spine have minimal stability requirements.

Possible Findings:

- FN: Ability to touch the inferior angle, no pain
 - » If thorax is normal bilaterally assume a postural and/or shoulder girdle stability/motor control dysfunction. If there is a thorax extension/rotation mobility dysfunction, complete your local biomechanical exam and treat the thorax.
- FP: Ability to touch the inferior angle, pain
 - » Proceed to the Passive Prone UE Pattern 1 Test.
- DN: Inability to touch the inferior angle, no pain
 - » Proceed to the Passive Prone UE Pattern 1 Test.
- DP: Inability to touch the inferior angle, pain
 - » Proceed to the Passive Prone UE Pattern 1 Test.



UPPER EXTREMITY BREAKOUTS



Test Objective for the Passive Prone Upper Extremity Pattern 1 Test:

To test passive shoulder mobility or stability/motor control in an unloaded posture.

How to Perform the Passive Prone Upper Extremity Pattern 1 Test:

Instruct the patient to assume the prone position with the arms and hands by the thighs. Passively move the patient's shoulder through UE Pattern 1 looking for their fingers to touch the opposite inferior angle of their scapula.

Functional Criteria:

Normal motion is fingertips reaching the opposite inferior angle of the scapula, without excessive scapular winging, without radial deviation, and/or without anterior dumping of the shoulder.

Additional Information:

Passively should be able to complete the motion with minimal torso rotation.

Possible Findings:

- FN: Ability to passively touch the inferior angle, no pain
 - » If thorax is normal bilaterally assume a postural and/or shoulder girdle stability/motor control dysfunction. If there is a thorax extension/rotation mobility dysfunction, complete your local biomechanical exam and treat the thorax.
- FP: Ability to passively touch the inferior angle, pain
 - » Proceed to Active Prone Shoulder 90/90 Internal Rotation Test.
- DN: Inability to passively touch the inferior angle, no pain
 - » Proceed to Active Prone Shoulder 90/90 Internal Rotation Test.
- DP: Inability to passively touch the inferior angle, pain
 - » Proceed to Active Prone Shoulder 90/90 Internal Rotation Test.



UPPER EXTREMITY BREAKOUTS



Test Objective for the Active Prone Shoulder 90/90 IR Test:

To test active shoulder internal rotation mobility and stability/motor control in an unloaded posture.

How to Perform the Active Prone Shoulder 90/90 IR Test:

Instruct the patient to assume the prone position with the testing arm hanging off the side of a table at approximately 30 degrees of horizontal adduction in a 90/90 position (90 degrees of shoulder abduction and 90 degrees of elbow flexion). Then have the patient try to internally rotate from the shoulder as far as possible.

Functional Criteria:

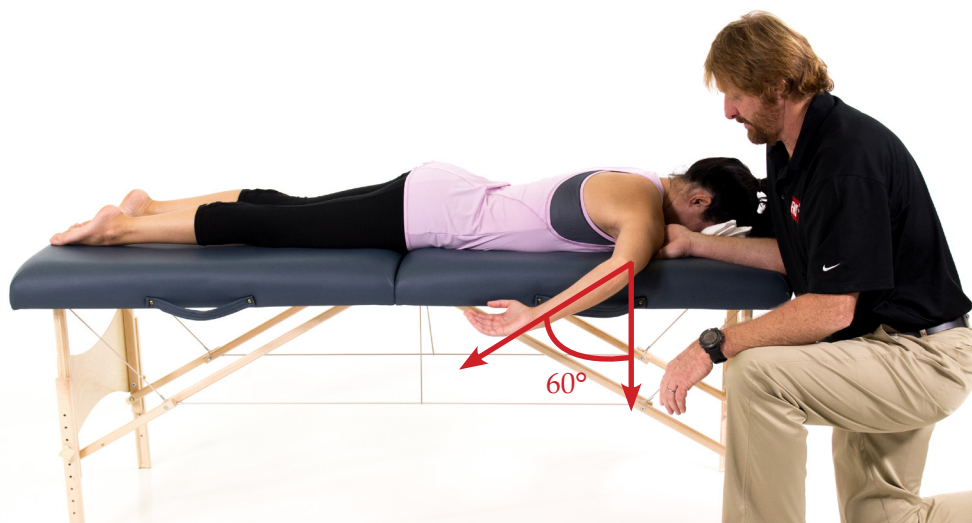
Using their forearm as the reference line, normal shoulder internal rotation is 60 degrees.

Additional Information:

If limited, take into consideration a total arc of motion - total arc of motion for the shoulder is 150 degrees. Calculate total arc of the shoulder by adding internal rotation degrees to external rotation degrees. If they add to 150 degrees or greater, assume the shoulder has normal mobility.

Possible Findings:

- FN: 60 degrees or greater of active shoulder internal rotation, no pain
 - » Consider shoulder internal rotation normal. Proceed to the Active Prone Shoulder Extension Test.
- FP: 60 degrees or greater of active shoulder internal rotation, pain
 - » Proceed to the Passive Prone Shoulder 90/90 IR Test.
- DN: Less than 60 degrees of active shoulder internal rotation and total arc less than 150 degrees, no pain
 - » Proceed to the Passive Prone Shoulder 90/90 IR Test.
- DP: Less than 60 degrees of active shoulder internal rotation and total arc less than 150 degrees, pain
 - » Proceed to the Passive Prone Shoulder 90/90 IR Test.



UPPER EXTREMITY BREAKOUTS



Test Objective for the Passive Prone Shoulder 90/90 IR Test:

To test passive shoulder internal rotation in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Prone Shoulder 90/90 IR Test:

Instruct the patient to assume the prone position with the testing arm hanging off the side of a table at approximately 30 degrees of horizontal adduction in a 90/90 position (90 degrees of shoulder abduction and 90 degrees of elbow flexion). Passively internally rotate from the shoulder as far as possible.

Functional Criteria:

Using their forearm as the reference line, normal shoulder internal rotation is 60 degrees.

Additional Information:

If limited, calculate total arc of the shoulder by adding internal rotation degrees to external rotation degrees. If total arc is 150 degrees or greater, assume the shoulder has normal mobility.

Possible Findings:

- FN: 60 degrees or greater of passive shoulder internal rotation, no pain
 - » Indicates a shoulder internal rotation stability/motor control dysfunction. Proceed to Active Prone Shoulder Extension Test.
- FP: 60 degrees or greater of passive shoulder internal rotation, pain
 - » Indicates pain with shoulder internal rotation - complete local biomechanical exam. Proceed to Active Prone Shoulder Extension Test.
- DN: Less than 60 degrees of passive internal rotation and total arc less than 150 degrees, no pain
 - » Indicates a shoulder internal rotation mobility dysfunction - complete local biomechanical exam. Proceed to Active Prone Shoulder Extension Test.
- DP: Less than 60 degrees of passive internal rotation and total arc less than 150 degrees, pain
 - » Indicates a painful shoulder internal rotation mobility dysfunction - complete local biomechanical exam. Proceed to Active Prone Shoulder Extension Test.



UPPER EXTREMITY BREAKOUTS



Test Objective for the Active Prone Shoulder Extension Test:

To assess active shoulder extension mobility and stability/motor control in an unloaded posture.

How to Perform the Active Prone Shoulder Extension Test:

Instruct the patient to assume the prone position with the arms and hands by the thighs with palms facing up. The patient then attempts to extend their arm up as high as possible into full shoulder extension, without moving into abduction.

Functional Criteria:

Using the Humerus as the reference line, normal shoulder extension is 50 degrees.

Additional Information:

Make sure the patient keeps their torso in contact with the table (no thoracic extension or rotation). Monitor for any anterior dumping of the shoulder.

Possible Findings:

- FN: 50 degrees or greater of active shoulder extension, no pain
 - » Consider shoulder extension normal. Proceed to the Active Prone Elbow Flexion Test (Extended).
- FP: 50 degrees or greater of active shoulder extension, pain
 - » Proceed to the Passive Prone Shoulder Extension Test.
- DN: Less than 50 degrees of active shoulder extension, no pain
 - » Proceed to the Passive Prone Shoulder Extension Test.
- DP: Less than 50 degrees of active shoulder extension, pain
 - » Proceed to the Passive Prone Shoulder Extension Test.



UPPER EXTREMITY BREAKOUTS



Test Objective for the Passive Prone Shoulder Extension Test:

To assess passive shoulder extension in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Prone Shoulder Extension Test:

Instruct the patient to assume the prone position with the arms and hands by the thighs with palms facing up. Passively extend the patient's arm without abduction or rotation of the torso.

Functional Criteria:

Using the Humerus as the reference line, normal shoulder extension is 50 degrees.

Additional Information:

Make sure the patient keeps their torso in contact with the table (no thoracic extension or rotation). Monitor for any anterior dumping of the shoulder.

Possible Findings:

- FN: 50 degrees or greater of passive shoulder extension, no pain
 - » Indicates a shoulder extension stability/motor control dysfunction. Proceed to the Active Prone Elbow Flexion Test (Extended).
- FP: 50 degrees or greater of passive shoulder extension, pain
 - » Indicates pain with shoulder extension - complete local biomechanical exam. Proceed to the Active Prone Elbow Flexion Test (Extended).
- DN: Less than 50 degrees of passive shoulder extension, no pain
 - » Indicates a shoulder extension mobility dysfunction - complete local biomechanical exam. Proceed to the Active Prone Elbow Flexion Test (Extended)
- DP: Less than 50 degrees of passive shoulder extension, pain
 - » Indicates a painful shoulder extension mobility dysfunction - complete local biomechanical exam. Proceed to the Active Prone Elbow Flexion Test (Extended).



UPPER EXTREMITY BREAKOUTS



Test Objective for the Active Prone Elbow Flexion Test (Extended):

To assess active elbow flexion mobility and stability/motor control in an unloaded posture with the shoulder extended.

How to Perform the Active Prone Elbow Flexion Test (Extended):

Instruct the patient to assume the prone position with the arms and hands by their thighs. The patient then flexes their elbow and tries to touch their thumb to their shoulder, keeping their upper arm extended down by their side.

Functional Criteria:

Normal range of motion for elbow flexion is their thumb to the front of their shoulder.

Additional Information:

Look for full flexion of the elbow in the shoulder extended position.

Possible Findings:

- FN: Ability to actively touch the thumb to the shoulder, no pain
 - » If no previous findings consider this a combined pattern one dysfunction. If there are previous findings, consider the elbow normal and treat your findings.
- FP: Ability to actively touch the thumb to the shoulder, pain
 - » Proceed to the Prone Passive Elbow Flexion Test (Extended).
- DN: Inability to actively touch the thumb to the shoulder, no pain
 - » Proceed to the Prone Passive Elbow Flexion Test (Extended).
- DP: Inability to actively touch the thumb to the shoulder, pain
 - » Proceed to the Prone Passive Elbow Flexion Test (Extended).



UPPER EXTREMITY BREAKOUTS



Test Objective for the Passive Prone Elbow Flexion Test (Extended):

To assess passive elbow flexion in an unloaded posture with the shoulder extended to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Prone Elbow Flexion Test (Extended):

Instruct the patient to assume the prone position with the arms and hands by their thighs. Passively flex their elbow and touch their thumb to their shoulder keeping their arm extended down by their side.

Functional Criteria:

Normal range of motion for elbow flexion is their thumb to the front of their shoulder.

Additional Information:

Look for full flexion of the elbow in the shoulder extended position.

Possible Findings:

- FN: Ability to passively touch the thumb to the shoulder, no pain
 - » Indicates an elbow flexion stability/motor control dysfunction in the shoulder extended position.
- FP: Ability to passively touch the thumb to the shoulder, pain
 - » Indicates pain with elbow flexion in the shoulder extended position - complete local biomechanical exam.
- DN: Inability to passively touch the thumb to the shoulder, no pain
 - » Indicates an elbow flexion mobility dysfunction in the shoulder extended position - complete local biomechanical exam.
- DP: Inability to passively touch the thumb to the shoulder, pain
 - » Indicates a painful elbow flexion mobility dysfunction in the shoulder extended position - complete local biomechanical exam.



UPPER EXTREMITY BREAKOUTS



Test Objective for the Active Prone Upper Extremity Pattern 2 Test:

To test active shoulder mobility and stability/motor control in an unloaded posture.

How to Perform the Active Prone Upper Extremity Pattern 2 Test:

Instruct the patient to assume the prone position, with the arms and hands by the thighs. Instruct the patient to try and reach overhead trying to touch the spine of the opposite scapula, repeating the Upper Extremity Pattern Two test.

Functional Criteria:

Normal motion is fingertips reaching the opposite spine of the scapula, without torso rotation.

Additional Information:

In this position the thoracic spine, shoulder girdle and cervical spine have minimal stability requirements. The patient should be able to complete the pattern without significant thorax rotation.

Possible Findings:

- FN: Ability to touch the spine of the opposite scapula, no pain
 - » If thorax is normal bilaterally assume a postural and/or shoulder girdle stability/motor control dysfunction. If there is a thorax extension/rotation mobility dysfunction, complete your local biomechanical exam and treat the thorax.
- FP: Ability to touch the spine of the opposite scapula, pain
 - » Proceed to the Passive Prone UE Pattern 2 Test.
- DN: Inability to touch the spine of the opposite scapula, no pain
 - » Proceed to the Passive Prone UE Pattern 2 Test.
- DP: Inability to touch the spine of the opposite scapula, pain
 - » Proceed to the Passive Prone UE Pattern 2 Test.



UPPER EXTREMITY BREAKOUTS



Test Objective for the Passive Prone Upper Extremity Pattern 2 Test:

To test passive shoulder mobility or stability/motor control, or both, in an unloaded posture.

How to Perform the Passive Prone Upper Extremity Pattern 2 Test:

Instruct the patient to assume the prone position with the arms and hands by the thighs. Passively move the patient's shoulder through UE Pattern 2 looking for their fingers to touch the spine of the opposite scapula.

Functional Criteria:

Normal motion is fingertips reaching the opposite spine of the scapula, without torso rotation.

Additional Information:

Passively should be able to complete the motion with minimal torso rotation.

Possible Findings:

- FN: Ability to passively touch the spine of the opposite scapula, no pain
 - » If thorax is normal bilaterally assume a postural and/or shoulder girdle stability/motor control dysfunction. If there is a thorax extension/rotation mobility dysfunction, complete your local biomechanical exam and treat the thorax.
- FP: Ability to passively touch the spine of the opposite scapula, pain
 - » Proceed to the Active Prone Shoulder 90/90 External Rotation Test.
- DN: Inability to passively touch the spine of the opposite scapula, no pain
 - » Proceed to the Active Prone Shoulder 90/90 External Rotation Test.
- DP: Inability to passively touch the spine of the opposite scapula, pain
 - » Proceed to the Active Prone Shoulder 90/90 External Rotation Test.



UPPER EXTREMITY BREAKOUTS



Test Objective for the Active Prone Shoulder 90/90 ER Test:

To test active shoulder external rotation mobility and stability/motor control in an unloaded posture.

How to Perform the Active Prone Shoulder 90/90 ER Test:

Instruct the patient to assume the prone position with the testing arm hanging off the side of a table at approximately 30 degrees of horizontal adduction in a 90/90 position (90 degrees shoulder abduction and 90 degrees of elbow flexion). Then have the patient try to externally rotate from the shoulder as far as possible.

Functional Criteria:

Using the forearm as the reference line, normal shoulder external rotation is 90 degrees.

Additional Information:

If limited, take into consideration a total arc of motion - total arc of motion for the shoulder is 150 degrees. Calculate total arc of the shoulder by adding internal rotation degrees to external rotation degrees. If they add to 150 degrees or greater, assume the shoulder has normal mobility.

Possible Findings:

- FN: 90 degrees or greater of active shoulder external rotation, no pain
 - » Consider shoulder external rotation normal. Proceed to the Active Prone Shoulder Flexion/Abduction Test.
- FP: 90 degrees or greater of active shoulder external rotation, pain
 - » Proceed to Passive Prone Shoulder 90/90 ER Test.
- DN: Less than 90 degrees of active shoulder external rotation and total arc less than 150 degrees, no pain
 - » Proceed to Passive Prone Shoulder 90/90 ER Test.
- DP: Less than 90 degrees of active shoulder external rotation and total arc less than 150 degrees, pain
 - » Proceed to Passive Prone Shoulder 90/90 ER Test.



UPPER EXTREMITY BREAKOUTS



Test Objective for the Passive Prone Shoulder 90/90 ER Test:

To test passive shoulder external rotation in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Prone Shoulder 90/90 ER Test:

Instruct the patient to assume the prone position with the testing arm hanging off the side of a table at approximately 30 degrees of horizontal adduction in a 90/90 position (90 degrees of shoulder abduction and 90 degrees of elbow flexion). Passively externally rotate from the shoulder as far as possible.

Functional Criteria:

Using the forearm as the reference line, normal shoulder external rotation is 90 degrees.

Additional Information:

If limited, take into consideration a total arc of motion - total arc of motion for the shoulder is 150 degrees. Calculate total arc of the shoulder by adding internal rotation degrees to external rotation degrees. If they add to 150 degrees or greater, assume the shoulder has normal mobility.

Possible Findings:

- FN: 90 degrees or greater of passive shoulder external rotation, no pain
 - » Indicates a shoulder external rotation stability/motor control dysfunction. Proceed to Active Prone Shoulder Flexion/Abduction Test.
- FP: 90 degrees or greater of passive shoulder external rotation, pain
 - » Indicates pain with shoulder external rotation - complete local biomechanical exam. Proceed to Active Prone Shoulder Flexion/Abduction Test.
- DN: Less than 90 degrees of passive shoulder external rotation and total arc less than 150 degrees, no pain
 - » Indicates a shoulder external rotation mobility dysfunction - complete local biomechanical exam. Proceed to Active Prone Shoulder Flexion/Abduction Test.
- DP: Less than 90 degrees of passive shoulder external rotation and total arc less than 150 degrees, pain
 - » Indicates a painful shoulder external rotation mobility dysfunction - complete local biomechanical exam. Proceed to Active Prone Shoulder Flexion/Abduction Test.



UPPER EXTREMITY BREAKOUTS



Test Objective for the Active Prone Shoulder Flexion/Abduction Test:

To test active shoulder flexion and abduction mobility and stability/motor control in an unloaded posture.

How to Perform the Active Prone Shoulder Flexion/Abduction Test:

Instruct the patient to assume the prone position with the arms and hands hanging down off the table. The patient then attempts to flex and abduct their arm up as high as possible into full shoulder flexion/abduction.

Functional Criteria:

Using the Humerus as the reference line, normal range of motion for shoulder flexion/abduction is 170 degrees.

Additional Information:

Make sure the patient keeps their torso in contact with the table (no thoracic extension/rotation).

Possible Findings:

- FN: 170 degrees or greater of active shoulder flexion/abduction, no pain
 - » Consider shoulder flexion/abduction normal. Proceed to the Active Prone Elbow Flexion Test (Flexed).
- FP: 170 degrees or greater of active shoulder flexion/abduction, pain
 - » Proceed to the Passive Prone Shoulder Flexion/Abduction Test.
- DN: Less than 170 degrees of active shoulder flexion/abduction, no pain
 - » Proceed to the Passive Prone Shoulder Flexion/Abduction Test.
- DP: Less than 170 degrees of active shoulder flexion/abduction, pain
 - » Proceed to the Passive Prone Shoulder Flexion/Abduction Test.



UPPER EXTREMITY BREAKOUTS



Test Objective for the Passive Prone Shoulder Flexion/Abduction Test:

To test passive shoulder flexion and abduction in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Prone Shoulder Flexion/Abduction Test:

Instruct the patient to assume the prone position with the arms and hands hanging down off the table. Passively flex and abduct their arm up as high as possible into full shoulder flexion/abduction.

Functional Criteria:

Using the Humerus as the reference line, normal range of motion for shoulder flexion/abduction is 170 degrees.

Additional Information:

Make sure the patient keeps their torso in contact with the table (no thoracic extension/rotation).

Possible Findings:

- FN: 170 degrees or greater of passive shoulder flexion/abduction, no pain
 - » Indicates a shoulder flexion/abduction stability/motor control dysfunction. Proceed to the Active Prone Elbow Flexion Test (Flexed).
- FP: 170 degrees or greater of passive shoulder flexion/abduction, pain
 - » Indicates pain with shoulder flexion/abduction - complete local biomechanical exam. Proceed to the Active Prone Elbow Flexion Test (Flexed).
- DN: Less than 170 degrees of passive shoulder flexion/abduction, no pain
 - » Indicates a shoulder flexion/abduction mobility dysfunction - complete local biomechanical exam. Proceed to the Active Prone Elbow Flexion Test (Flexed).
- DP: Less than 170 degrees of passive shoulder flexion/abduction, pain
 - » Indicates a painful shoulder flexion/abduction mobility dysfunction - complete local biomechanical exam. Proceed to the Active Prone Elbow Flexion Test (Flexed).



UPPER EXTREMITY BREAKOUTS



Test Objective for the Active Prone Elbow Flexion Test (Flexed):

To test active elbow flexion mobility and stability/motor control in an unloaded posture with the shoulder flexed.

How to Perform the Active Prone Elbow Flexion Test (Flexed):

Instruct the patient to assume the prone position with the arms and hands by their thighs. The patient then takes the right arm into full shoulder flexion first and then flexes their elbow trying to touch their thumb to their shoulder, keeping their shoulder flexed throughout the movement.

Functional Criteria:

Normal range of motion for elbow flexion is their thumb to the front of their shoulder.

Additional Information:

They should be able to maintain their end range shoulder flexed/abducted position while flexing elbow - monitor position, but do not stabilize.

Possible Findings:

- FN: Ability to actively touch the thumb to the shoulder in the full shoulder flexed position, no pain
 - » If no previous findings, consider this a combined pattern two dysfunction. If there are previous findings, consider the elbow normal and treat your findings.
- FP: Ability to actively touch the thumb to the shoulder in the full shoulder flexed position, pain
 - » Proceed to the Passive Prone Elbow Flexion Test (Flexed).
- DN: Inability to actively touch the thumb to the shoulder in the full shoulder flexed position, no pain
 - » Proceed to the Passive Prone Elbow Flexion Test (Flexed).
- DP: Inability to actively touch the thumb to the shoulder in the full shoulder flexed position, pain
 - » Proceed to the Passive Prone Elbow Flexion Test (Flexed).



UPPER EXTREMITY BREAKOUTS



Test Objective for the Passive Prone Elbow Flexion Test (Flexed):

To test passive elbow flexion in an unloaded posture with the shoulder flexed to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Prone Elbow Flexion Test (Flexed):

Instruct the patient to assume the prone position with the arms and hands by their thighs. Passively flex their arm and try to passively flex their elbow, trying to touch their thumb to their shoulder - keeping the arm flexed.

Functional Criteria:

Normal range of motion for elbow flexion is their thumb to the front of their shoulder.

Additional Information:

You should be able to maintain their end range shoulder flexed/abducted position while flexing elbow.

Possible Findings:

- FN: Ability to passively touch the thumb to the shoulder, no pain
 - » Indicates an elbow flexion stability/motor control dysfunction in the shoulder flexed position.
- FP: Ability to passively touch the thumb to the shoulder, pain
 - » Indicates pain with elbow flexion in the shoulder flexed position - complete local biomechanical exam.
- DN: Inability to passively touch the thumb to the shoulder, no pain
 - » Indicates an elbow flexion mobility dysfunction in the shoulder flexed position - complete local biomechanical exam.
- DP: Inability to passively touch the thumb to the shoulder, pain
 - » Indicates a painful elbow flexion mobility dysfunction in the shoulder flexed position - complete local biomechanical exam.

